Citi Bike Rebalancing Explorer · Product Reference

What it is, how it works, and where it stops

Live app: nyc-citibike-rebalancing.vercel.app · **Code and data:** github.com/rezataeb/nyc-citibike-rebalancing
Data: trips Oct 2025 – Sep 2026; station availability Jul 13, 2026 – Oct 31, 2026. Independent project, public data only. Not an official NYC DOT or Lyft product.
About the numbers: every figure here is read from the committed data files when this page is built (`python3 -m report.build_report`), not typed in.

1. Main goal

An empty or full bike-share station is a barrier to choosing bike share, not only an operations problem. The tool shows **where and when** Citi Bike stations empty or fill, **how reliable** they really were, **who lives near** the worst ones (NYCHA housing, schools, subway gaps), and **what a truck fleet** could realistically fix. It runs in the browser on precomputed files, at $0 to host and $0 per visit.

2. Users

User	The question they bring	Where they find the answer
DOT planners and analysts	Where does imbalance concentrate, and who does it affect?	Historical Flow map, Equity context, Reliability
Operations and oversight staff	Where should a truck go first? What can a fleet cover?	Priority Truck Stops, fleet simulator
Council and community board staff	How does the station on my block behave?	Ask the data: a plain-English briefing per station
Riders and the public	Is my station usually empty at 8 AM? Is it empty right now?	Station card, Live Docks

3. Main features

Historical Flow

Net flow (arrivals minus departures) for 2,523 stations, by hour, weekday or weekend, season and month. An hour slider replays the day; a ribbon totals the bikes out of place.

Live Docks

Bikes and open docks right now, from the public GBFS feed, refreshed every 60 seconds. An amber ring marks a station that is empty or full while its neighbors within 500 m are too (the project's own approximation of the contract's adjacency rule, not a penalty calculation).

Reliability

How often each station had 0 bikes or 0 docks in the logged snapshots, with its rank among the 2,507 logged stations, and what station size is associated with.

Priority Truck Stops and the fleet simulator

A ranked morning route (pick up here, drop off there), with an optional equity weight; a slider runs 1 to 10 trucks.

Scenario Planner

Weather presets (steady rain, snow, ideal riding weather, a hot day) projected from past weather effects and labeled PROJECTED; equity distance sliders; the fleet slider. Each reads a precomputed file next to the baseline.

Ask the data (AI)

Type a station for a two-to-three sentence briefing, written in advance by Claude Haiku 5.5 from that station's own record (2,430 stations). Ready-made questions follow the New York clock ("Where do bikes run out next hour?", "Where do docks fill up next hour?", "Which stations are empty right now?"); these are computed by code, not AI.

4. Dataset

Layer	Source	Coverage
Trips	Citi Bike S3 trip archives	12 months (Oct 2025 – Sep 2026): the baseline year; 2,523 stations, 2,054 with trips every month
Station status	Citi Bike GBFS feed	1,222 snapshots by an hourly GitHub Action, Jul 13, 2026 – Oct 31, 2026; 2,922,701 station observations. Also read live.
NYCHA, schools	NYC Open Data	216 developments; 1,899 schools (2019–2020 locations)
Subway	NY State Open Data	2,120 station records
Weather	Open-Meteo archive	Daily temperature and precipitation, 18 zones on a 6 km grid

Baseline year, not a deadline. Twelve months hold one of each calendar month, so "a typical November weekday at 9 AM" has an answer on any date. The window is set in one file (`pipeline/window.py`) and one command rebuilds everything; the availability window is frozen at its end date, so later log files never change a published rate.

5. From raw data to the screen

1. **Acquire.** Download the monthly trip archives; log the GBFS feed every hour to a separate `data-snapshots` branch; fetch the Open Data layers and the weather archive once.
2. **Clean.** Drop trips under 60 seconds or over 4 hours. Merge station ids that were split across months (for example `X.10` and `X.1`) and drop rows outside the window. Exclude offline stations from every failure denominator.
3. **Aggregate.** Count arrivals and departures per station, day and hour; average into weekday and weekend curves per season and month (`flows.json`). Hours without trips count as zero.

4. **Enrich.** Distance from each station to the nearest NYCHA development, school and subway station; flag 300 m (NYCHA, school) and 800 m (subway gap).
5. **Model.** Cluster station types; fit weather effects; compare forecasts by walk-forward and forward-only tests; fit capacity against reliability (Section 6).
6. **Predict and simulate.** Weather scenarios from the fitted effects; ten truck-fleet scenarios from the morning deficit and surplus.
7. **Explain.** Build a facts record per station; the model writes a briefing from it; code checks every number and claim, retries once, then falls back to a template.
8. **Check and publish.** Ground-truth spot checks (a Financial District station must fill in the morning, Thanksgiving must dip), the test suite, then a static deploy. No backend, no database.

6. Models behind it

Model	Method	Result
Station typology	K-means on each station's weekday hourly shape; k chosen by silhouette	k = 2 (silhouette 0.41): 694 commuter core, 1,115 residential feeder; 714 too quiet to cluster
Demand forecast	Seasonal naive vs. spline GAM vs. gradient boosting with an extrapolation guard; 12-fold walk-forward test, Wilcoxon signed-rank	Naive best: MAE 1.92 vs. 2.03 guarded ($p = 0.012$); naive better in 9 of 12 months
Forward-only benchmark	Profile, last week, four-week mean, Holt-Winters (weekly season), gradient boosting with lags and weather; 38 weekly folds, 2,054 stations, zero-filled	Profile best hourly and daily; for weekday 6–9 AM totals Holt-Winters is +16.6% (34 of 38 weeks)
Weather effects	Elasticities of daily flow to temperature and precipitation	Drive the four weather presets; a heat-wave preset was left out (the fit does not reach that far)
Reliability	Share of online observations with 0 bikes or 0 docks	13.1% unusable; full outnumbers empty 2.8 to 1; median station 10.8%
Capacity vs. reliability	OLS: unusable rate ~ capacity + log(1 + demand), 2,438 stations	40+ docks about 2.2 points lower than under 20 (a pattern, not proof)
Fleet simulator	Greedy routing; 20 bikes per truck, at most 45 stops	10 trucks fully rebalance 155 of 1,039 deficit stations (14.9%); the stop cap binds
Station briefings	Claude Haiku 5.5 via the Batch API, one facts record per call; code checker; template fallback	2,430 briefings; 98.6% AI-written, 34 from the template

7. Main decisions, and why

- **Baseline first.** Every model is compared with a plain baseline, and the baseline won, so the map shows historical patterns as the main view and labels the comparison openly.

- **Honest tests.** Walk-forward folds, not one split; a second benchmark that is forward-only and counts zero hours, because the first could learn from later months.
- **Precompute, then serve static files.** No model runs in the browser and nothing needs a server, so hosting is free and every number can be traced to a file in the repo.
- **The pipeline computes; the language model only explains.** Briefings are written once, checked by code (a number not in the record is rejected), and shipped as JSON. No API key in the browser, no cost per visit. Ranked answers (the ready-made questions) are computed by code because a ranking must be exact.
- **One window, set in one place.** A test blocks dates typed into the pages; every label is built from the data.
- **Label what is not measured.** PROJECTED on weather scenarios; "a pattern, not proof" on capacity; the 500 m adjacency is called the project's own choice.
- **Scope kept small on purpose.** No map-wide equity filter (equity stays per station); a dock-capacity sandbox was closed because public data never recorded capacity over time.

8. Limitations

- **Sampling.** Availability is logged about hourly (median gap 65 minutes) with a 14.5-day gap (Aug 30, 2026 – Sep 13, 2026). Rates are estimates; outage durations and the contract's 60/120-minute thresholds cannot be measured.
- **Short availability window.** 110 days, summer into fall; no seasonal comparison yet.
- **Correlation only.** The capacity result compares different stations; borough and zone are not controlled for.
- **Projections.** Weather scenarios extrapolate past effects; the truck route starts at a stand-in depot and uses a fixed stop cap, not a time or distance budget.
- **Forecasting.** The better morning-peak forecast (Holt-Winters) is tested but not yet shown in the dashboard.
- **AI briefings.** They restate a station's own numbers; they do not explain causes. The school layer dates from 2019–2020.

Public data: Citi Bike GBFS and trip data, NYC Open Data, NY State Open Data, Open-Meteo. Rebuild with `python3 -m report.build_report`. Full detail: the project report (same folder).